

LECTURE 10

ENDOGENEITY AND INSTRUMENTAL VARIABLES ESTIMATION

Endogeneity

Consider the partitioned regression model:

$$\begin{aligned} Y_i &= X_i^\top \beta + U_i \\ &= X_{1i}^\top \beta_1 + X_{2i}^\top \beta_2 + U_i, \end{aligned} \quad (1)$$

where X_{1i} is a k_1 -vector and X_{2i} is a k_2 -vector of random regressors, β_1 and β_2 are $k_1 \times 1$ and $k_2 \times 1$ vectors of unknown parameters, respectively, with $k_1 + k_2 = k$. We assume that X_{1i} is *endogenous*:

$$E[X_{1i}U_i] \neq 0,$$

as opposed to the (*weakly*) *exogenous* regressors X_{2i} :

$$E[X_{2i}U_i] = 0.$$

(The assumption $E[U_i | X_{2i}] = 0$ is called *strong exogeneity*.) Sources of endogeneity:

- **Omitted variables.** Consider the wage equation:

$$\begin{aligned} \log(\text{Wage}_i) &= \alpha + \beta \text{Education}_i + \gamma \text{Gender}_i + \delta \text{Ability}_i + V_i \\ &= \alpha + \beta \text{Education}_i + \gamma \text{Gender}_i + U_i. \end{aligned}$$

Since ability is unobservable, it enters the error term $U_i = \delta \text{Ability}_i + V_i$. The gender variable is plausibly exogenous; however, education is correlated with ability and therefore endogenous.

- **Errors in variables.** Suppose that the true model is

$$Y_i = \tilde{X}_{1i}^\top \beta + X_{2i}^\top \beta_2 + V_i;$$

however, $\tilde{X}_{1i}$ is unobservable. Instead, the econometrician observes $X_{1i} = \tilde{X}_{1i} + \varepsilon_i$, where ε_i is some noise vector independent of $\tilde{X}_{1i}$ and X_{2i} . Substituting $\tilde{X}_{1i} = X_{1i} - \varepsilon_i$ into the above equation,

$$Y_i = X_{1i}^\top \beta + X_{2i}^\top \beta_2 - \varepsilon_i^\top \beta + V_i.$$

Set $U_i = -\varepsilon_i^\top \beta + V_i$. While X_{2i} is exogenous, X_{1i} is endogenous, because it is correlated with U_i through ε_i .

- **Simultaneity.** Consider the following equation:

$$\text{Hours}_i = \beta_1 \text{Children}_i + X_{2i}^\top \beta_2 + U_i,$$

where Hours_i is the hours of work per week, Children_i is the number of children in the family, and X_{2i} is a vector of exogenous variables. While the number of children affects labor supply, it is reasonable to assume that career decisions affect family size. Accordingly, there is another equation determining the number of children in the family:

$$\text{Children}_i = \gamma_1 \text{Hours}_i + Z_{1i}^\top \gamma_2 + V_i,$$

where Z_{1i} is another vector of exogenous variables. Substituting the expression for the hours into the equation for the number of children, we obtain (assuming that $1 - \beta_1 \gamma_1 \neq 0$)

$$\text{Children}_i = X_{2i}^\top \left(\frac{\beta_2 \gamma_1}{1 - \beta_1 \gamma_1} \right) + Z_{1i}^\top \frac{\gamma_2}{1 - \beta_1 \gamma_1} + \frac{\gamma_1}{1 - \beta_1 \gamma_1} U_i + \frac{1}{1 - \beta_1 \gamma_1} V_i.$$

Assuming that X_{2i} , Z_{1i} , and V_i are uncorrelated with U_i ,

$$\begin{aligned} E[U_i \text{Children}_i] &= \frac{\gamma_1}{1 - \beta_1 \gamma_1} E[U_i^2] \\ &\neq 0. \end{aligned}$$

Properties of the OLS under endogeneity

Consider first the OLS estimator of β_1 :

$$\begin{aligned}\hat{\beta}_{1n} &= (X_1^\top M_2 X_1)^{-1} X_1^\top M_2 Y \\ &= \beta_1 + (X_1^\top M_2 X_1)^{-1} X_1^\top M_2 U,\end{aligned}$$

where $M_2 = I_n - X_2 (X_2^\top X_2)^{-1} X_2^\top$. Then

$$\begin{aligned}n^{-1} X_1^\top M_2 X_1 &= n^{-1} \sum_{i=1}^n X_{1i} X_{1i}^\top - n^{-1} \sum_{i=1}^n X_{1i} X_{2i}^\top \left(n^{-1} \sum_{i=1}^n X_{2i} X_{2i}^\top \right)^{-1} n^{-1} \sum_{i=1}^n X_{2i} X_{1i}^\top, \\ n^{-1} X_1^\top M_2 U &= n^{-1} \sum_{i=1}^n X_{1i} U_i - n^{-1} \sum_{i=1}^n X_{1i} X_{2i}^\top \left(n^{-1} \sum_{i=1}^n X_{2i} X_{2i}^\top \right)^{-1} n^{-1} \sum_{i=1}^n X_{2i} U_i.\end{aligned}$$

Assume that:

- $\{(Y_i, X_i) : i \geq 1\}$ are iid.
- $E[X_{i,j}^2] < \infty$ for all $j = 1, \dots, k$.
- $E[X_i X_i^\top]$ is positive definite.
- $E[U_i^2] < \infty$.

By the WLLN, we have

$$\begin{aligned}n^{-1} \sum_{i=1}^n X_{1i} X_{1i}^\top &\rightarrow_p E[X_{1i} X_{1i}^\top], \\ n^{-1} \sum_{i=1}^n X_{1i} X_{2i}^\top &\rightarrow_p E[X_{1i} X_{2i}^\top], \\ n^{-1} \sum_{i=1}^n X_{2i} X_{2i}^\top &\rightarrow_p E[X_{2i} X_{2i}^\top], \\ n^{-1} \sum_{i=1}^n X_{2i} U_i &\rightarrow_p 0, \\ n^{-1} \sum_{i=1}^n X_{1i} U_i &\rightarrow_p E[X_{1i} U_i].\end{aligned}$$

Thus,

$$\begin{aligned}n^{-1} X_1^\top M_2 X_1 &\rightarrow_p E[X_{1i} X_{1i}^\top] - E[X_{1i} X_{2i}^\top] (E[X_{2i} X_{2i}^\top])^{-1} E[X_{2i} X_{1i}^\top], \\ n^{-1} X_1^\top M_2 U &\rightarrow_p E[X_{1i} U_i] - E[X_{1i} X_{2i}^\top] (E[X_{2i} X_{2i}^\top])^{-1} E[X_{2i} U_i] \\ &= E[X_{1i} U_i] \\ &\neq 0,\end{aligned}$$

and we conclude that $\hat{\beta}_{1n}$ is inconsistent:

$$\begin{aligned}\hat{\beta}_{1n} &\rightarrow_p \beta_1 + \left(E[X_{1i} X_{1i}^\top] - E[X_{1i} X_{2i}^\top] (E[X_{2i} X_{2i}^\top])^{-1} E[X_{2i} X_{1i}^\top] \right)^{-1} E[X_{1i} U_i] \\ &\neq \beta_1.\end{aligned}$$

The OLS estimator of β_2 is also inconsistent, as the following argument shows. We have

$$\hat{\beta}_{2n} = \beta_2 + (X_2^\top M_1 X_2)^{-1} X_2^\top M_1 U,$$

where $M_1 = I_n - X_1 (X_1^\top X_1)^{-1} X_1^\top$. Then

$$\begin{aligned} \hat{\beta}_{2n} &\rightarrow_p \beta_2 - \left(E[X_{2i} X_{2i}^\top] - E[X_{2i} X_{1i}^\top] (E[X_{1i} X_{1i}^\top])^{-1} E[X_{1i} X_{2i}^\top] \right)^{-1} E[X_{2i} X_{1i}^\top] (E[X_{1i} X_{1i}^\top])^{-1} E[X_{1i} U_i] \\ &\neq \beta_2. \end{aligned}$$

Instrumental Variables Estimation

Let Z_{1i} be a k_1 -vector of *exogenous* variables:

$$E[Z_{1i} U_i] = 0.$$

It is important that Z_{1i} is *excluded* from the model (1), i.e., Z_{1i} does not contain any of the elements of X_{2i} . Define

$$\begin{aligned} X_i &= \begin{pmatrix} X_{1i} \\ X_{2i} \end{pmatrix}, \\ Z_i &= \begin{pmatrix} Z_{1i} \\ X_{2i} \end{pmatrix}. \end{aligned}$$

Here X_i is the k -vector of regressors and Z_i is the k -vector of *instrumental variables* (IVs). The exogenous regressors appear again in the vector of IVs, and for each endogenous regressor, we include an exogenous variable (IV) that must be excluded from the model $Y_i = X_i^\top \beta + U_i$. When all regressors are endogenous ($k_1 = k$), X_i and Z_i share no elements.

We assume that the IVs are informative about the regressors. This is expressed as the following *rank condition*:

$$\text{rank}(E[Z_i X_i^\top]) = k. \quad (2)$$

The rank condition in (2) will fail if, for example, $E[Z_{1i} X_i^\top] = 0$ (Z_{1i} is exogenous but merely random noise). The rank condition will also fail if some of the elements of Z_{1i} are linear combinations of the elements of the included exogenous regressors X_{2i} .

Example. Consider the Hours/Children example. Angrist and Evans (1998) suggested using the sex composition of the first two children as an instrument for the number of children in the family (the sample was restricted to women with at least two children). This is motivated by the observation that if the first two children are of the same sex (boy-boy or girl-girl), the family is more likely to have a third child than in the opposite-sex case (boy-girl or girl-boy), so the dummy variable indicating same-sex siblings is positively correlated with the total number of children. On the other hand, the instrument is uncorrelated with the errors, because sex composition is determined randomly.

Since

$$E[Z_i U_i] = 0,$$

the MM principle suggests an estimator that solves the following system of k equations:

$$\begin{aligned} n^{-1} \sum_{i=1}^n Z_i \left(Y_i - X_i^\top \hat{\beta}_n^{IV} \right) &= 0, \text{ or} \\ \hat{\beta}_n^{IV} &= \left(\sum_{i=1}^n Z_i X_i^\top \right)^{-1} \sum_{i=1}^n Z_i Y_i \\ &= (Z^\top X)^{-1} Z^\top Y. \end{aligned}$$

The estimator $\widehat{\beta}_n^{IV}$ is called the IV estimator of β .

Next, we show consistency and asymptotic normality of the IV estimator. We assume:

- $\{(Y_i, X_i, Z_i) : i \geq 1\}$ are iid.
- $E[Z_i U_i] = 0$.
- $E[X_{i,j}^2] < \infty$ for all $j = 1, \dots, k$.
- $E[Z_{i,j}^2] < \infty$ for all $j = 1, \dots, k$.
- $E[Z_i X_i^\top]$ is of rank k .
- $E[U_i^2 Z_i Z_i^\top]$ is positive definite.

Write

$$\widehat{\beta}_n^{IV} = \beta + \left(n^{-1} \sum_{i=1}^n Z_i X_i^\top \right)^{-1} n^{-1} \sum_{i=1}^n Z_i U_i. \quad (3)$$

By the Cauchy-Schwarz inequality,

$$\begin{aligned} E |Z_{i,r} X_{i,s}| &\leq \sqrt{E[Z_{i,r}^2] E[X_{i,s}^2]} \\ &< \infty \text{ for all } r, s = 1, \dots, k. \end{aligned}$$

Therefore, by Slutsky's Theorem,

$$\begin{aligned} \widehat{\beta}_n^{IV} &\rightarrow_p \beta + (E[Z_i X_i^\top])^{-1} E[Z_i U_i] \\ &= \beta. \end{aligned}$$

To show asymptotic normality, we assume in addition that

- $E[Z_{i,j}^4] < \infty$ for all $j = 1, \dots, k$.
- $E[U_i^4] < \infty$.

Write (3) as

$$n^{1/2} (\widehat{\beta}_n^{IV} - \beta) = \left(n^{-1} \sum_{i=1}^n Z_i X_i^\top \right)^{-1} n^{-1/2} \sum_{i=1}^n Z_i U_i.$$

As in Lecture 7, for all $r, s = 1, \dots, k$,

$$\begin{aligned} E |U_i^2 Z_{i,r} Z_{i,s}| &\leq (E[U_i^4])^{1/2} (E[Z_{i,r}^4] E[Z_{i,s}^4])^{1/4} \\ &< \infty. \end{aligned}$$

By the CLT and Cramér convergence theorem,

$$\begin{aligned} n^{1/2} (\widehat{\beta}_n^{IV} - \beta) &\rightarrow_d (E[Z_i X_i^\top])^{-1} N(0, E[U_i^2 Z_i Z_i^\top]) \\ &= N(0, (E[Z_i X_i^\top])^{-1} E[U_i^2 Z_i Z_i^\top] (E[X_i Z_i^\top])^{-1}). \end{aligned}$$

The asymptotic covariance matrix takes the sandwich form and can be estimated consistently by

$$\left(n^{-1} \sum_{i=1}^n Z_i X_i^\top \right)^{-1} n^{-1} \sum_{i=1}^n \widehat{U}_i^2 Z_i Z_i^\top \left(n^{-1} \sum_{i=1}^n X_i Z_i^\top \right)^{-1},$$

where $\widehat{U}_i = Y_i - X_i^\top \widehat{\beta}_n^{IV}$.

Weak Instruments

Consider the following regression model with a single (endogenous) regressor:

$$Y_i = \beta X_i + U_i.$$

We assume that the endogenous regressor is related to a single IV through the following equation:

$$\begin{aligned} X_i &= \pi_n Z_i + V_i, \\ \pi_n &= \frac{c}{\sqrt{n}}. \end{aligned}$$

This parameterization captures a weak but nonzero correlation between X_i and its instrument Z_i . We will rely on a large- n approximation for the distribution of the estimators, and, therefore, weakness of the relationship between X and Z has to be modeled in terms of the sample size n . This is because any fixed $\pi \neq 0$ will be “large” when $n \rightarrow \infty$. Therefore, we assume that $\pi_n \rightarrow 0$ as $n \rightarrow \infty$. The rate of convergence is chosen in such a way that small correlations captured by the nonzero constant c will appear in the limit. In addition, we assume that the IV is uncorrelated with the errors U and V , and that the errors are homoskedastic (conditional on Z). The homoskedasticity assumption is not essential and is made only for simplicity.

- $\{(Y_i, X_i, Z_i) : i = 1, \dots, n\}$ are iid.
- $E \left[Z_i \begin{pmatrix} U_i \\ V_i \end{pmatrix} \right] = 0$.
- $E \left[\begin{pmatrix} U_i \\ V_i \end{pmatrix} \begin{pmatrix} U_i \\ V_i \end{pmatrix}^\top \mid Z_i \right] = \begin{pmatrix} \sigma_U^2 & \sigma_{UV} \\ \sigma_{UV} & \sigma_V^2 \end{pmatrix} = \Sigma$, a positive definite matrix.
- $E[Z_i^2] = Q < \infty$.

Under the above assumptions,

$$\begin{aligned} n^{-1} \sum_{i=1}^n Z_i^2 &\rightarrow_p Q, \\ n^{-1/2} \sum_{i=1}^n Z_i \begin{pmatrix} U_i \\ V_i \end{pmatrix} &\rightarrow_d N \left(0, Q \begin{pmatrix} \sigma_U^2 & \sigma_{UV} \\ \sigma_{UV} & \sigma_V^2 \end{pmatrix} \right) \\ &\equiv \begin{pmatrix} \Psi_U \\ \Psi_V \end{pmatrix}. \end{aligned}$$

These results follow from the WLLN and the CLT, respectively. In the result above, Ψ_U and Ψ_V are jointly normal random variables with zero means, covariance $\sigma_{UV}Q$, and variances $\sigma_U^2 Q$ and $\sigma_V^2 Q$, respectively.

Next, the IV estimator of β is given by

$$\begin{aligned} \hat{\beta}_n^{IV} - \beta &= \frac{\sum_{i=1}^n Z_i Y_i}{\sum_{i=1}^n Z_i X_i} - \beta \\ &= \frac{\sum_{i=1}^n Z_i U_i}{\sum_{i=1}^n Z_i (\pi_n Z_i + V_i)} \\ &= \frac{\sum_{i=1}^n Z_i U_i}{cn^{-1/2} \sum_{i=1}^n Z_i^2 + \sum_{i=1}^n Z_i V_i} \\ &= \frac{n^{-1/2} \sum_{i=1}^n Z_i U_i}{cn^{-1} \sum_{i=1}^n Z_i^2 + n^{-1/2} \sum_{i=1}^n Z_i V_i} \\ &\rightarrow_d \frac{\Psi_U}{cQ + \Psi_V}, \end{aligned}$$

where the last result follows by the CMT and the joint convergence in distribution of $n^{-1/2} \sum_{i=1}^n Z_i U_i$ and $n^{-1/2} \sum_{i=1}^n Z_i V_i$. Hence, due to the weak IV problem, the IV estimator of β is not consistent, and converges instead to a random limit. Furthermore, due to the inconsistency of $\hat{\beta}_n^{IV}$, confidence intervals and hypothesis tests based on $\hat{\beta}_n^{IV}$ are invalid: the asymptotic coverage probabilities and size are incorrect. Moreover, the asymptotic distribution of $\hat{\beta}_n^{IV}$ depends on unknown parameters c and Σ that cannot be consistently estimated under these circumstances. Hence, the distribution of $\hat{\beta}_n^{IV} - \beta$ cannot be evaluated.

Although consistent point estimation is impossible in this setting, hypotheses about β can still be tested. This can be done with the so-called Anderson-Rubin (AR) statistic. Consider the null hypothesis $H_0 : \beta = \beta_0$. The null-restricted residuals are given by

$$U_{0,i} = Y_i - \beta_0 X_i.$$

Provided that the null hypothesis is true, we have the true residuals, and therefore

$$\begin{aligned} n^{-1} \sum_{i=1}^n U_{0,i}^2 &\rightarrow_p \sigma_U^2, \\ n^{-1/2} \sum_{i=1}^n Z_i U_{0,i} &\rightarrow_d \Psi_U \\ &\stackrel{d}{=} N(0, \sigma_U^2 Q). \end{aligned}$$

The AR statistic is given by

$$AR_n(\beta_0) = \frac{n^{-1/2} \sum_{i=1}^n Z_i (Y_i - \beta_0 X_i)}{\sqrt{(n^{-1} \sum_{i=1}^n U_{0,i}^2) (n^{-1} \sum_{i=1}^n Z_i^2)}}.$$

Under the null hypothesis,

$$AR_n(\beta_0) \rightarrow_d N(0, 1).$$

(The statistic above is a simplified version of the AR statistic. The actual AR statistic is designed for models with more than one IV, also accommodates exogenous regressors, and is therefore constructed as a quadratic form with a chi-squared asymptotic distribution under the null.) One should reject the null in favor of the alternative, say $H_1 : \beta \neq \beta_0$, if $|AR_n(\beta_0)| > z_{1-\alpha/2}$, where z_τ is the τ quantile of the standard normal distribution. Such a test will have asymptotic size α regardless of the strength of the IV. Furthermore, one can construct confidence intervals for β by collecting all values β_0 for which the AR test cannot reject the null:

$$CI_{1-\alpha,n} = \{\beta_0 : |AR_n(\beta_0)| \leq z_{1-\alpha/2}\}.$$

If the null hypothesis is false, then $U_{0,i}$'s are not the true residuals, but a function of X_i 's and the residuals:

$$U_{0,i} = U_i + (\beta - \beta_0) X_i.$$

If the IV Z_i is related to the regressor X_i , i.e., $c \neq 0$, then the asymptotic distribution of the AR statistic will be different from the standard normal, and the test will have power to reject the null. The power of the test depends on the distance from the null $|\beta - \beta_0|$ and the strength of the IV, which is captured by c . If $c = 0$, i.e., the IV is irrelevant, then the AR test has no power to reject false null hypotheses. The test will reject the null with probability α , equal to the size, regardless of the value of β (the power function is flat and equal to the size, α , for all values of β). In this case, the confidence interval described above will have infinite length (it will include all values $\beta_0 \in \mathbb{R}$).

The approach can be extended to the case of IV regression with multiple endogenous and exogenous regressors, as well as heteroskedastic errors. However, in the case of weak IVs, in general, one cannot test hypotheses on subvectors of β using the AR test. Because the coefficient estimators are inconsistent, obtaining true null-restricted residuals requires specifying the values of all coefficients under the null.